

MGT3003

Customer Relationship Management Project Report

FORD MOTOR COMPANY

**ANALYZING MODERN CRM PRACTICES AND THEIR IMPACT
ON CUSTOMER RETENTION: A CASE STUDY OF FORD
MOTOR COMPANY**

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CHAPTER 1: INTRODUCTION TO MODERN CRM PRACTICES

1.1 Overview of Modern CRM Practices and Their Importance

Customer Relationship Management (CRM) is a strategic approach that organizations use to manage interactions with current and potential customers. It involves the integration of people, processes, and technology to understand customer behavior, improve customer satisfaction, and ultimately drive business growth. In today's highly competitive and digital-driven business environment, CRM has evolved from a simple data management tool into a comprehensive system that supports customer-centric strategies.

Modern CRM practices emphasize building long-term relationships rather than focusing solely on short-term transactions. Organizations now leverage CRM to gain deep insights into customer preferences, behaviors, and expectations. This allows companies to deliver personalized experiences, which are crucial for maintaining a competitive edge in the market.

With the advancement of technologies such as Artificial Intelligence (AI), Big Data analytics, and cloud computing, CRM systems have become more sophisticated. These technologies enable businesses to analyze vast amounts of customer data in real time, predict customer needs, and offer tailored solutions. For instance, predictive analytics helps companies identify potential churn risks and take proactive measures to retain customers.

Another key aspect of modern CRM is omni-channel engagement, where businesses interact with customers through multiple channels such as websites, mobile applications, social media, email, and physical stores. This ensures a seamless and consistent customer experience across all touchpoints. Companies like Ford integrate digital platforms such as mobile apps and connected vehicle technologies to enhance customer interaction and engagement.

CRM also plays a critical role in improving operational efficiency. By automating routine tasks such as marketing campaigns, customer support, and follow-ups, organizations can focus more on strategic decision-making. Furthermore, CRM systems facilitate better communication and collaboration among different departments, ensuring a unified approach toward customer management.

In today's business landscape, CRM is not just an optional tool but a necessity. It helps organizations build strong customer relationships, increase customer satisfaction, and achieve sustainable growth. Companies that effectively implement CRM strategies are more likely to succeed in retaining customers and enhancing their brand value.

1.2 Importance of CRM in Today's Business Environment

The importance of CRM has significantly increased due to changing customer expectations and the rapid growth of digital technologies. Customers today expect personalized experience, quick responses, and seamless interactions across various platforms. CRM systems enable organizations to meet these expectations effectively.

One of the primary benefits of CRM is enhanced customer engagement. By analyzing customer data, companies can understand individual preferences and tailor their offerings accordingly. This leads to improved customer satisfaction and stronger relationships. For example, Ford uses CRM systems to personalize communication and provide proactive services, such as maintenance reminders and customized offers. CRM also contributes to data-driven decision-making. Organizations can use customer data to identify trends, evaluate performance, and make informed business decisions. This reduces uncertainty and increases the effectiveness of marketing and sales strategies.

Another significant advantage is improved customer retention. Acquiring new customers is often more expensive than retaining existing ones. CRM helps businesses identify loyal customers, understand their needs, and provide value-added services to keep them engaged. This not only reduces churn rates but also increases customer lifetime value.

Additionally, CRM enhances cross-functional collaboration within an organization. Departments such as marketing, sales, and customer service can access shared customer data, ensuring consistent communication and better coordination. This leads to more cohesive customer experience. In summary, CRM is a powerful tool that enables organizations to build strong customer relationships, improve operational efficiency, and achieve long-term success in a competitive market.

1.3 Key CRM Concepts

Modern CRM practices are built on several fundamental concepts that guide organizations in managing customer relationships effectively. These concepts are essential for understanding how CRM contributes to business success.

1.3.1 Customer Retention

Customer retention refers to the ability of a company to retain its existing customers over a period. It involves strategies aimed at keeping customers engaged and satisfied, thereby reducing the likelihood of them switching to competitors. Retention strategies include personalized communication, loyalty programs, and excellent customer service. High retention rates are indicative of strong customer relationships and contribute significantly to long-term profitability.

1.3.2 Customer Loyalty

Customer loyalty is the commitment of customers to continue purchasing products or services from a particular brand. Loyal customers are more likely to make repeat purchases and recommend the brand to others. CRM systems help build loyalty by offering personalized experiences, rewards programs, and consistent value. For instance, loyalty programs like reward points encourage customers to stay connected with the brand.

1.3.3 Customer Equity

Customer equity refers to the total value of all customer relationships a company maintains. It is considered a key metric for assessing the long-term health and profitability of a business. By focusing on customer equity, organizations shift their attention from individual transactions to the overall value of their customer base. CRM systems play a crucial role in measuring and enhancing customer equity through data analysis and targeted strategies.

1.3.4 Relationship Marketing

Relationship marketing emphasizes building long-term relationships with customers rather than focusing on one-time sales. It involves continuous engagement, personalized communication, and value creation. This approach helps organizations establish trust and foster emotional connections with customers.

1.3.5 Customer Lifetime Value (CLV)

Customer Lifetime Value (CLV) is the total revenue a business can expect from a customer throughout their relationship. CRM systems use data analytics to estimate CLV and identify high-value customers. This allows organizations to allocate resources effectively and focus on strategies that maximize long-term profitability. In conclusion, modern CRM practices have transformed the way organizations interact with their customers. By leveraging advanced technologies and data-driven insights, businesses can deliver personalized experiences, improve customer satisfaction, and build long-term relationships. Key concepts such as customer retention, loyalty, and equity play a vital role in shaping effective CRM strategies.

As businesses continue to evolve in a digital-first environment, the importance of CRM will only increase. Organizations that prioritize customer relationships and invest in CRM technologies are better positioned to achieve sustainable growth and maintain a competitive advantage

CHAPTER 2: COMPANY PROFILE – FORD MOTOR COMPANY

2.1 Background and History of Ford Motor Company

Ford Motor Company, founded in 1903 by Henry Ford, is one of the oldest and most influential automobile manufacturers in the world. Headquarters in Dearborn, Michigan, Ford revolutionized the automotive industry by introducing mass production techniques, making automobiles affordable and accessible to the public. In its early years, Ford focused on producing reliable and affordable vehicles, which helped it build a strong customer base. The introduction of the assembly line significantly reduced production costs and increased efficiency, setting new standards in manufacturing.

Over the decades, Ford expanded its operations globally, establishing a strong presence in various international markets. The company adapted its strategies to meet the diverse needs of customers across different regions, emphasizing localized customer engagement.

In recent years, Ford has undergone a significant transformation, shifting from a traditional automaker to a customer-focused mobility company. This transition is driven by advancements in technology and changing consumer preferences. The company now focuses on delivering innovative, sustainable, and connected mobility solutions.

2.2 Key Milestones, Awards, and Recognitions

Ford Motor Company's long-standing legacy in the automotive industry is marked by numerous milestones, innovations, and recognitions that reflect its commitment to excellence, technological advancement, and customer-centric strategies. These milestones not only highlight Ford's growth as an organization but also demonstrate how it has continuously adapted to changing market conditions and customer expectations.

2.2.1 Early Milestones and Industrial Innovation

One of the most significant milestones in Ford's history was the introduction of the Model T in 1908, which revolutionized the automobile industry. The Model T was designed to be affordable, durable, and easy to maintain, making car ownership accessible to the public. This innovation transformed transportation and established Ford as a pioneer in mass production.

Another groundbreaking achievement was the implementation of the assembly line manufacturing system in 1913. This innovation significantly reduced production time and costs, allowing Ford to produce vehicles at an unprecedented scale. The assembly line not only improved efficiency but also set new standards for industrial manufacturing worldwide. These early milestones laid the foundation for Ford's customer-focused philosophy by emphasizing affordability, accessibility, and reliability—principles that continue to influence its CRM strategies today.

2.2.2 Expansion and Global Recognition

As Ford grew, it expanded its operations beyond the United States, establishing a strong global presence. The company entered international markets across Europe, Asia, and other regions, adapting its products and strategies to meet diverse customer needs.

During this period, Ford achieved several milestones, including:

- Establishing manufacturing plants worldwide
- Introducing region-specific vehicle models
- Building a strong global dealer network

This expansion enabled Ford to strengthen its customer relationships by understanding local market preferences and delivering tailored solutions. The company's ability to localize its offerings played a crucial role in enhancing customer satisfaction and loyalty on a global scale.

2.2.3 Technological Advancements and Product Innovation

Ford has consistently been at the forefront of technological innovation in the automotive industry. Over the years, the company has introduced several advanced features and technologies that have improved vehicle performance, safety, and customer experience.

Some key technological milestones include:

- Development of advanced safety systems and driver assistance technologies
- Introduction of fuel-efficient and environmentally friendly vehicles
- Investment in electric and hybrid vehicle technologies

In recent years, Ford has made significant progress in the electric vehicle (EV) segment with models such as the Mustang Mach-E and the F-150 Lightning. These innovations demonstrate Ford's commitment to sustainability and future mobility. Additionally, Ford's integration of digital technologies, such as connected vehicle systems and over-the-air (OTA) updates, has enhanced the overall customer experience. These advancements are closely linked to CRM, as they enable continuous interaction and engagement with customers even after the purchase.

2.2.4 CRM Evolution and Digital Transformation Milestones

Ford's journey in Customer Relationship Management has evolved significantly over the years. Initially, the company relied on dealer-based systems to manage customer relationships. However, with the advancement of technology, Ford transitioned to more integrated and data-driven CRM systems.

Key CRM-related milestones include:

- **Early Dealer CRM Systems:** Focused on localized customer interactions and relationship building
- **Enterprise-wide CRM Integration:** Unified customer data across departments and touchpoints
- **Launch of FordPass Ecosystem:** A digital platform that provides connected services, rewards, and personalized experiences
- **Adoption of AI and Analytics:** Use of predictive models to understand customer behavior and improve engagement

The introduction of the FordPass platform marked a major milestone in Ford's CRM strategy. It enabled customers to interact with their vehicles and the brand through a single digital interface, enhancing convenience and engagement. These developments reflect Ford's shift from a transactional approach to a relationship-driven model, where customer experience is at the center of all operations.

2.2.5 Awards and Industry Recognitions

Ford has received numerous awards and recognitions for its excellence in innovation, quality, safety, and customer satisfaction. These accolades highlight the company's leadership in the automotive industry and its commitment to delivering value to customers.

Some notable areas of recognition include:

- **Customer Satisfaction Awards:** Ford has been recognized for its efforts in improving customer experience and service quality
- **Innovation Awards:** Acknowledgment for advancements in vehicle technology, digital platforms, and sustainability initiatives
- **Safety Ratings:** High safety ratings for various vehicle models, reflecting Ford's focus on customer well-being
- **Sustainability Recognition:** Awards for environmental initiatives and development of electric vehicles

In addition, Ford's digital transformation and CRM initiatives have been acknowledged within the industry. The company's use of data analytics, AI, and connected technologies has set benchmarks for customer engagement and relationship management.

2.2.6 Continuous Investment in Customer Experience

A key factor behind Ford's success is its continuous investment in enhancing customer experience. The company actively develops new technologies and strategies to improve customer interaction and satisfaction.

Key initiatives include:

- Implementation of connected vehicle technologies for real-time data and services
- Development of personalized marketing strategies using CRM data
- Enhancement of after-sales services through digital platforms
- Introduction of loyalty programs to retain customers

These initiatives demonstrate Ford's commitment to building long-term relationships with customers and maximizing customer lifetime value.

2.2.7 Strategic Transformation and Future Milestones

Ford's recent transformation under the Ford+ strategy represents a significant milestone in its journey. The company is transitioning from a traditional automobile manufacturer to a mobility solutions provider, focusing on:

- Electric vehicles and sustainable mobility
- Digital ecosystems and connected services
- Customer-centric innovation

This strategic shift is expected to shape the future of Ford and redefine its relationship with customers. By integrating CRM with advanced technologies, Ford aims to deliver personalized and seamless experiences across all touchpoints.

2.3 Overview of the Company's Management Practices, Business Operations, Products, and Services

Ford Motor Company has evolved into a highly customer-centric organization where management practices, business operations, and product offerings are deeply integrated with Customer Relationship Management (CRM) strategies. This transformation reflects the company's shift from being a traditional automobile manufacturer to a modern mobility solutions provider focused on innovation, digitalization, and customer engagement. In today's competitive automotive industry, organizations must go beyond simply selling products; they must build long-term relationships with customers. Ford achieves this by embedding CRM into every aspect of its operations, from leadership decisions to customer service delivery. The company leverages advanced technologies, data analytics, and digital platforms to understand customer needs and deliver personalized experiences.

This section provides a comprehensive overview of Ford's management practices, operational strategies, and its wide range of products and services, highlighting how CRM acts as a unifying force across all these dimensions.

2.3.1 Management Practices at Ford

Customer-Centric Leadership

Ford's leadership is built on a strong customer-first philosophy, where customer satisfaction and long-term relationship building are prioritized over short-term profits. The company's executives emphasize the importance of delivering value at every stage of the customer journey.

The Ford strategy plays a crucial role in this leadership approach. It focuses on:

- Enhancing digital customer experiences
- Leveraging connected vehicle technology
- Providing personalized services
- Strengthening customer engagement

Leadership ensures that CRM is not treated as a standalone system but as a strategic capability that influences all business decisions. This approach helps Ford maintain a competitive advantage in a rapidly evolving market.

2.3.2 Strategic Alignment and Organizational Structure

Ford's organizational structure is designed to support both traditional and modern mobility solutions while maintaining a unified CRM approach.

The company operates in two major divisions:

- Ford Blue – Focused on internal combustion engine (ICE) vehicles
- Ford Model e – Focused on electric vehicles (EVs) and digital innovation

This strategic separation allows Ford to:

- Optimize performance in both segments
- Accelerate innovation in electric mobility
- Maintain operational efficiency

At the same time, CRM systems ensure that customer data and insights are shared across all divisions. This cross-functional alignment between marketing, sales, IT, and customer service ensures a consistent and seamless customer experience.

2.3.3 Data-Driven Decision Making

Ford relies heavily on data to guide its management practices. CRM systems collect and analyze customer data from multiple sources, including:

- Dealership interactions
- Mobile applications (FordPass)
- Websites and digital platforms
- Service and maintenance records

Using advanced analytics and artificial intelligence, Ford can:

- Predict customer preferences and behavior
- Identify potential churn risks
- Segment customers for targeted marketing
- Improve product design and innovation

This data-driven approach enables Ford to make informed decisions, reduce uncertainties, and enhance customer satisfaction.

2.3.4 Dealer Network Integration

Ford's global dealer network is a key component of its CRM ecosystem. Dealers serve as the primary interface between the company and its customers, providing personalized services and maintaining long-term relationships.

CRM systems enable dealers to access real-time customer information, allowing them to:

- Understand customer preferences and history
- Provide personalized recommendations
- Deliver consistent service quality

This integration ensures that customers receive a seamless experience regardless of location, strengthening brand trust and loyalty.

2.3.5 Feedback and Continuous Improvement

Ford places significant emphasis on continuous improvement through customer feedback. The company uses various mechanisms to collect feedback, including:

- Voice of Customer (VoC) programs
- Customer satisfaction surveys
- Social media monitoring
- Service feedback systems

This feedback is analyzed to identify gaps and improve products, services, and customer interactions. Continuous improvement ensures that Ford remains responsive to changing customer expectations and maintains high levels of satisfaction.

2.3.6 Business Operations at Ford

Integrated Customer Journey Management

Ford's business operations are designed to manage the entire customer lifecycle, from initial awareness to post-purchase engagement. CRM systems track every interaction, ensuring a smooth and personalized experience.

The customer journey includes:

1. Awareness and research
2. Vehicle selection and purchase
3. Ownership and maintenance
4. Loyalty and repeat purchases

By managing this end-to-end journey, Ford enhances customer satisfaction and builds long-term relationships.

Omni-Channel Engagement

Ford adopts an omni-channel strategy to engage customers across multiple platforms, including:

- Physical dealerships
- Official websites
- Mobile applications (FordPass)
- Social media platforms

This approach ensures that customers can interact with the brand seamlessly across different channels. For example, a customer may explore vehicle options online and finalize the purchase at a dealership.

Automation and Digital Communication

Automation plays a vital role in improving efficiency and customer engagement. Ford uses CRM systems to automate:

- Email marketing campaigns
- Service reminders and notifications
- Promotional offers
- Customer support interactions

These communications are personalized based on customer data, ensuring relevance and effectiveness.

Connected Vehicle Ecosystem

Ford integrates connected vehicle technology into its operations, enabling real-time data collection and analysis. Vehicles equipped with telematics systems provide valuable insights that help:

- Monitor vehicle performance
- Send maintenance alerts
- Offer personalized driving insights

This proactive approach enhances customer experience and strengthens trust in the brand.

Predictive Analytics and CRM Integration

Predictive analytics are used to anticipate customer needs and improve operational efficiency. By analyzing historical data, Ford can:

- Identify customers at risk of leaving
- Predict service and maintenance requirements
- Optimize marketing strategies

This integration of analytics with CRM enables Ford to proactively address customer needs and reduce churn rates.

2.3.7 Products Offered by Ford

Diverse Vehicle Portfolio

Ford offers a wide range of vehicles to meet the needs of different customer segments, including:

- Passenger cars
- SUV'S
- Trucks (e.g., F-Series)
- Electric vehicles (EVs)

CRM systems help manage customer interactions across all product categories, ensuring personalized experiences.

Electric Vehicles and Innovation

Ford is actively investing in electric vehicles and sustainable mobility solutions. Popular EV models include:

- Mustang Mach-E
- F-150 Lightning

CRM supports EV customers by providing:

- Charging support information
- Range management tools
- Over-the-air (OTA) updates

This enhances customer convenience and satisfaction.

Commercial and Fleet Solutions

Ford provides specialized solutions for business customers, including fleet management services. These solutions are supported by CRM systems that help manage large-scale operations and maintain strong business relationships.

2.3.8 Services Provided by Ford

Digital Platforms (FordPass Ecosystem)

FordPass is a key digital platform that enhances customer engagement by offering:

- Remote vehicle access
- Service scheduling
- Rewards programs
- Real-time vehicle insights

After-Sales Services

After-sales services are critical for customer retention and satisfaction. These include:

- Maintenance and repairs
- Warranty services
- Customer support

CRM ensures timely communication and efficient service delivery.

Financial Services

Ford provides financial solutions such as:

- Vehicle loans
- Leasing options
- Insurance services

These services are integrated into CRM systems for seamless customer experience.

Loyalty Programs

Ford's loyalty programs, such as FordPass Rewards, encourage repeat purchases and long-term engagement by offering incentives and benefits.

Connected and Personalized Services

Using CRM data, Ford delivers personalized services such as:

- Customized offers
- Driving insights
- Proactive maintenance alerts

These services enhance customer satisfaction and strengthen relationships.

Integration of CRM Across Management, Operations, and Services

CRM serves as the backbone of Ford's entire business ecosystem. It connects:

- Management strategies
- Business operations
- Product offerings
- Customer services

This integration ensures consistency, efficiency, and a superior customer experience across all touchpoints.

CHAPTER 3: CRM PRACTICES IN THE FIRM

3.1 Introduction to CRM in the Modern Business Context

Customer Relationship Management (CRM) has evolved from a basic data storage system into a comprehensive strategic framework that enables organizations to build, maintain, and enhance long-term relationships with customers. In today's highly competitive and digital-driven environment, companies are no longer focused solely on acquiring customers but are increasingly prioritizing customer retention, loyalty, and lifetime value. CRM systems play a critical role in achieving these objectives by integrating customer data, analysing behaviour, and enabling personalized interactions.

Ford Motor Company represents a strong example of this transformation, as it has shifted from a traditional automotive manufacturer to a customer-centric mobility organization. This shift is driven by the understanding that customer relationships extend far beyond the point of sale and continue throughout the entire lifecycle of product ownership. Through the implementation of advanced CRM practices, Ford aims to enhance customer engagement, improve satisfaction, and ultimately maximize long-term profitability.

3.2 CRM Strategy and Digital Transformation at Ford

Ford's CRM strategy is deeply integrated into its broader digital transformation initiative known as the "Ford+ Plan." This strategy emphasizes the importance of leveraging digital technologies, connected vehicle data, and artificial intelligence to create a seamless and personalized customer experience.

One of the key components of Ford's CRM strategy is omni-channel engagement. Customers interact with the brand through multiple touchpoints, including dealerships, websites, mobile applications, and social media platforms. Ford ensures that these interactions are interconnected, allowing for a consistent and unified customer journey. For instance, a customer may begin their journey by researching a vehicle online, continue through a dealership visit, and later interact with the company via the FordPass mobile application.

Another significant aspect of Ford's CRM strategy is the use of connected vehicle technology. Modern Ford vehicles are equipped with sensors and telematics systems that generate real-time data about vehicle performance and customer usage patterns. This data is utilized to provide proactive services such as maintenance alerts, personalized recommendations, and usage-based services. Such proactive engagement enhances customer satisfaction and strengthens trust in the brand.

3.3 Exploratory Data Analysis and Customer Profiling

The exploratory data analysis (Figure.1) conducted on the CRM dataset provides valuable insights into customer demographics and behaviour. The dataset includes variables such as age, income, purchase frequency, purchase amount, email interactions, and click behaviour, all of which contribute to understanding customer engagement patterns.

The distribution of age indicates a widespread across different age groups, suggesting that Ford caters to a diverse customer base. Income distribution shows a slight skew toward higher values, indicating the presence of premium customer segments that may contribute significantly to revenue generation. Purchase frequency and purchase amount provide insights into customer activity levels, with higher values indicating stronger engagement and loyalty.



Figure.1. Exploratory Data Analysis

From Figure.2, The analysis of churn status reveals that a significant proportion of customers have discontinued their relationship with the company. This highlights the importance of identifying factors that contribute to churn and implementing strategies to reduce it. Additionally, the distribution of customers across different locations indicates that urban areas have the highest concentration of customers, followed by suburban and rural regions.

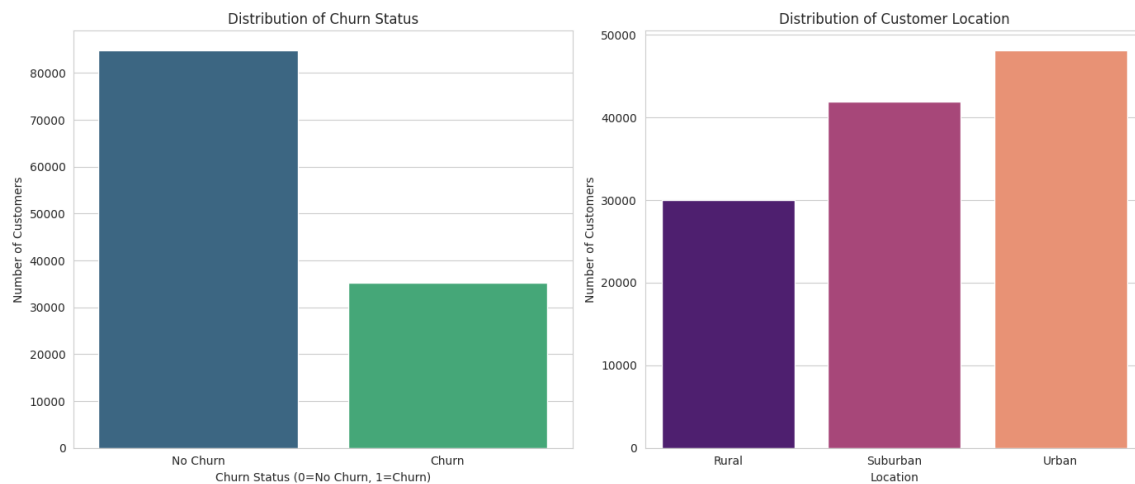


Figure.2. (a). Distribution of Churn Status (b). Distribution of Customer Location

3.4 Correlation Analysis and Key Drivers of Churn

From Figure.3, Correlation analysis plays a crucial role in identifying the relationships between different variables and their impact on customer churn. The correlation matrix reveals that purchase frequency has the strongest negative correlation with churn, indicating that customers who engage more frequently with the company are less likely to leave.

Income also exhibits a slight negative correlation with churn, suggesting that higher-income customers tend to remain loyal to the brand. On the other hand, variables such as email opens and clicks show minimal correlation with churn, indicating that engagement alone is not sufficient to ensure retention.

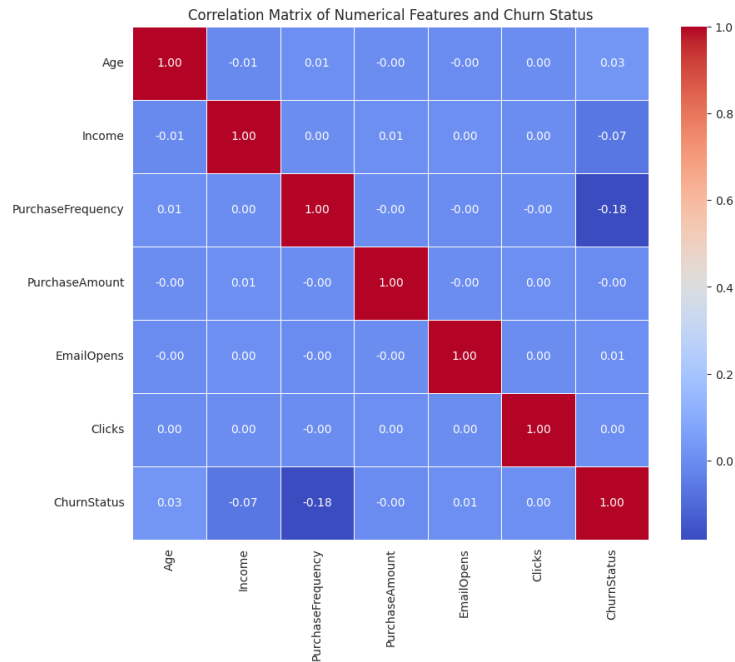


Figure.3. Correlation Matrix of Numerical Features and Churn Status

These findings emphasize the importance of focusing on meaningful interactions and value-driven engagement rather than relying solely on superficial metrics.

3.5 Predictive Modelling for Churn Identification

From Figure.4, To better understand and predict customer churn, logistic regression was applied as a baseline model. The model achieved an accuracy of approximately 70%, indicating moderate predictive performance. However, a deeper analysis of the classification report reveals that the model is biased toward predicting non-churn cases, resulting in poor detection of actual churn instances.

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...
--- Logistic Regression for Churn Factors ---

Logistic Regression Model Accuracy: 0.7064

Classification Report:
              precision    recall  f1-score   support

     0       0.71         1.00         0.83     25436
     1       0.49         0.01         0.02     10564

 accuracy         0.71         0.71         0.71     36000
 macro avg       0.60         0.50         0.42     36000
 weighted avg    0.64         0.71         0.59     36000

```

Figure.4. Predictive Modelling for Churn Identification

This limitation highlights the challenges associated with imbalanced datasets and underscores the need for advanced modelling techniques. Despite these limitations, logistic regression provides valuable insights into the relative importance of different variables and serves as a foundation for further analysis.

3.6 Customer Segmentation and Behavioral Clustering

Customer segmentation is a critical component of CRM, as it enables organizations to tailor their strategies to different customer groups. In this study, clustering techniques were used to segment customers based on their demographic and behavioral attributes. The elbow method (Figure.5) was applied to determine the optimal number of clusters, which was found to be three. This suggests that the customer base can be effectively divided into three distinct groups with unique characteristics.

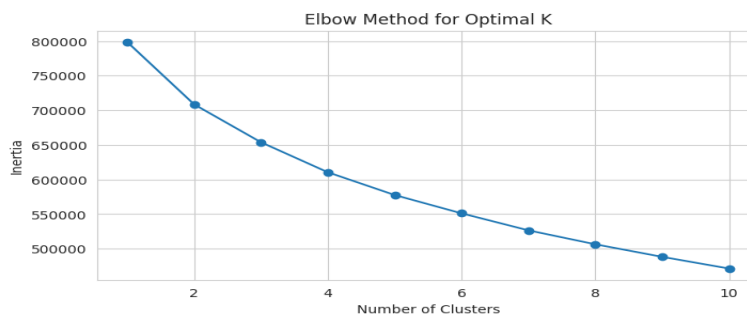


Figure.5. Elbow Method

The scatter plot of clusters based on age and income (Figure.6) reveals overlapping regions, indicating that segmentation based solely on demographics may not be sufficient. Instead, behavioral factors such as purchase frequency and engagement play a more significant role in defining customer segment.



Figure.6. Customer Segments based on Income and Age

Further analysis of purchase patterns across clusters highlights differences in spending behaviour, with certain clusters demonstrating higher purchase amounts and greater engagement levels.

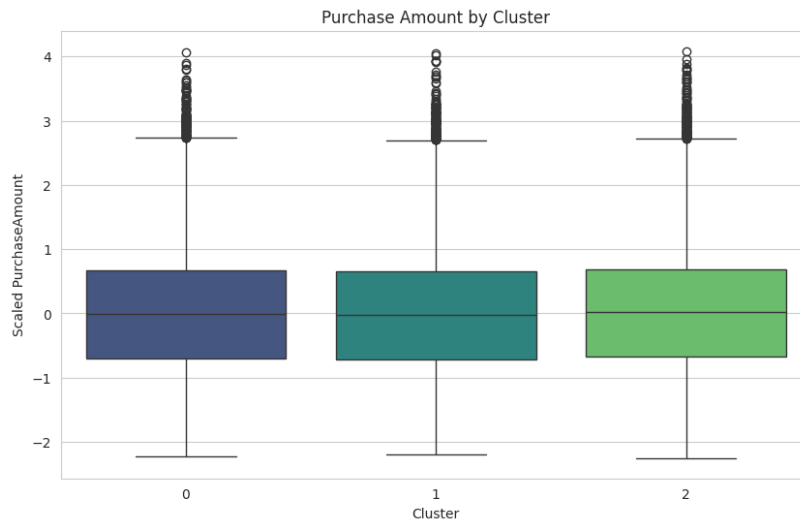


Figure.7. Purchase Amount by Cluster

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--- Cluster Characteristics ---
...

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	Age	Income	PurchaseFrequency	PurchaseAmount	EmailOpens	\
Cluster						
0	-0.003044	-0.005075	0.928852	-0.007675	-0.010274	
1	-0.008066	-0.004866	-0.932806	-0.017498	-0.002724	
2	0.009265	0.008323	-0.009841	0.021004	0.010934	

	Clicks	ChurnStatus
Cluster		
0	-0.624938	0.214461
1	-0.634671	0.371815
2	1.054396	0.295106

	Location_Urban	Location_Suburban	Location_Rural
Cluster			
0	0.404028	0.349912	0.246060
1	0.402878	0.349719	0.247402
2	0.396305	0.348748	0.254947

Figure.8. Cluster Characteristics

3.7 Customer Lifetime Value and Retention Analysis

Customer Lifetime Value (CLV) is a key metric that represents the total value a customer generates over the course of their relationship with the company. In this study, CLV was analysed using Kaplan-Meier survival analysis, which provides insights into customer retention over time.

The survival curve indicates a gradual decline in the proportion of active customers, with approximately 60% of customers retained after one year. This suggests that while a significant number of customers remain loyal, there is also a considerable drop-off that needs to be addressed.

--- Survival Analysis for Customer Lifetime Value (CLV) ---

Kaplan-Meier Survival Function Summary:

timeline	KM_estimate
0.0	0.999250
1.0	0.998292
2.0	0.997600
3.0	0.996750
4.0	0.995983

Figure.9. Kaplan-Meier Survival Summary

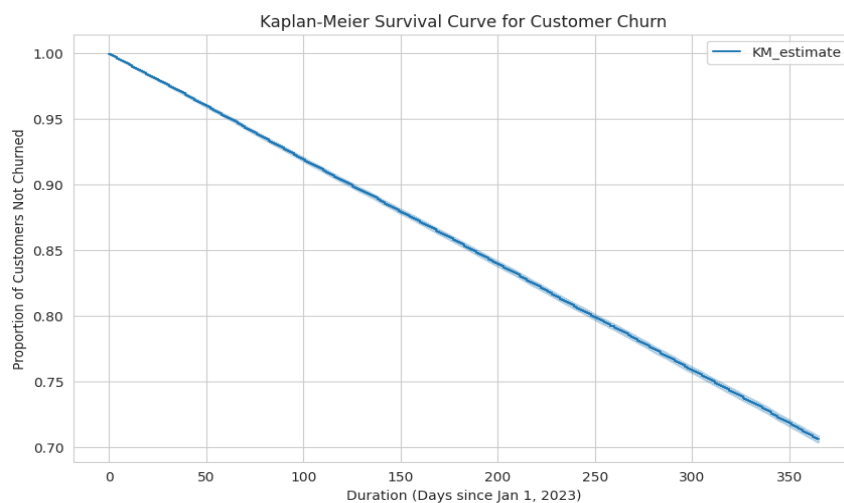


Figure.10. Kaplan-Meier Survival Curve for Customer Retention

The analysis further reveals that the initial phase of the customer lifecycle is critical, as most churn occurs during this period. Customers who remain engaged beyond this stage are more likely to continue their relationship with the company. Location-based analysis shows minimal variation in retention rates, indicating that geographic factors do not significantly influence customer behaviour.

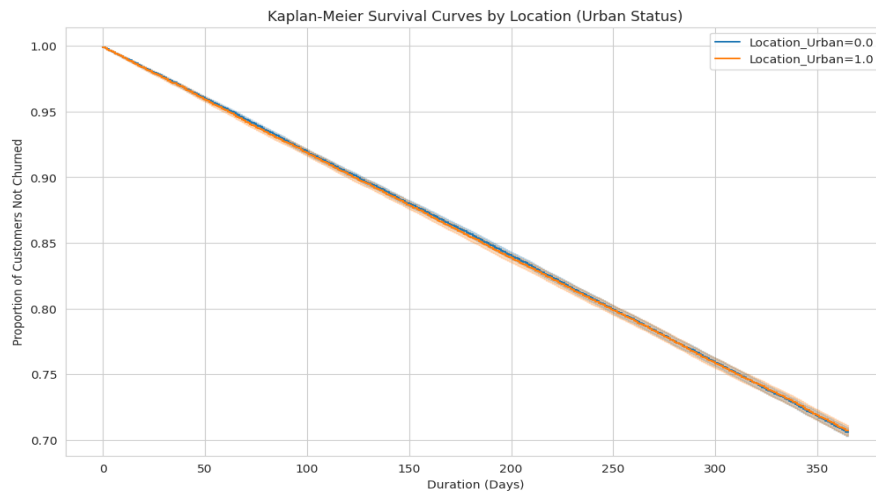


Figure.11. Survival Curve by Customer Location

3.8 Marketing Campaign Performance Analysis

Marketing campaigns play an important role in influencing customer behaviour and driving engagement. The analysis of purchase amounts across different campaigns reveals slight variations in performance, with some campaigns achieving higher average purchase values.

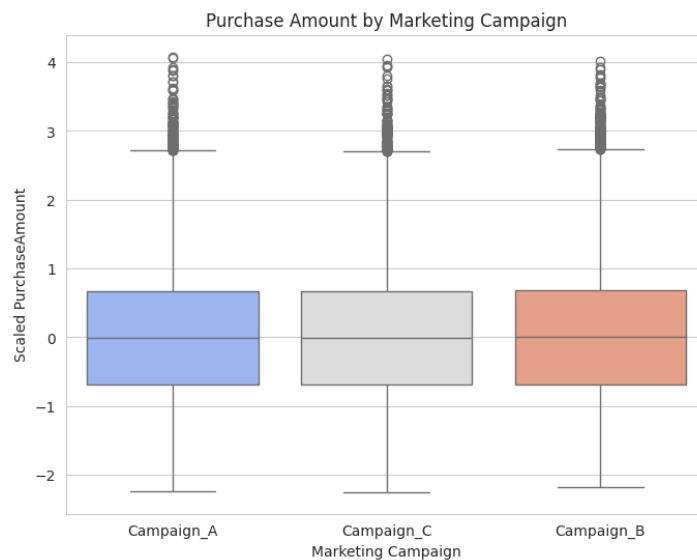


Figure.12. Purchase Amount by Marketing Campaign

These findings suggest that while marketing efforts are effective, there is potential for further optimization through personalization and targeted strategies.

3.9 Advanced Machine Learning Models for Churn Prediction

To improve predictive performance, advanced machine learning models such as Random Forest and Decision Tree were implemented. The Random Forest model achieved similar accuracy to logistic regression but provided additional insights through feature importance analysis.

```
. --- Random Forest Classifier for Churn Prediction ---

Random Forest Model Accuracy: 0.6995
Random Forest Model Precision: 0.4047
Random Forest Model Recall: 0.0510
Random Forest Model F1-Score: 0.0906
Random Forest Model ROC AUC Score: 0.5868

Classification Report:
              precision    recall  f1-score   support

     0         0.71         0.97         0.82        25436
     1         0.40         0.05         0.09        10564

 accuracy         0.56         0.51         0.46        36000
  macro avg         0.56         0.51         0.46        36000
 weighted avg         0.62         0.70         0.61        36000
```

Figure.13. Random Forest Model Performance

```
Top 10 Random Forest Feature Importances:
Income                0.205651
PurchaseAmount        0.198754
EmailOpens            0.165472
Clicks                0.147925
Age                   0.140718
PurchaseFrequency     0.099299
Location_Urban        0.014756
Location_Suburban     0.014379
Location_Rural        0.013045
```

Figure.14. Random Forest Feature Importance Values

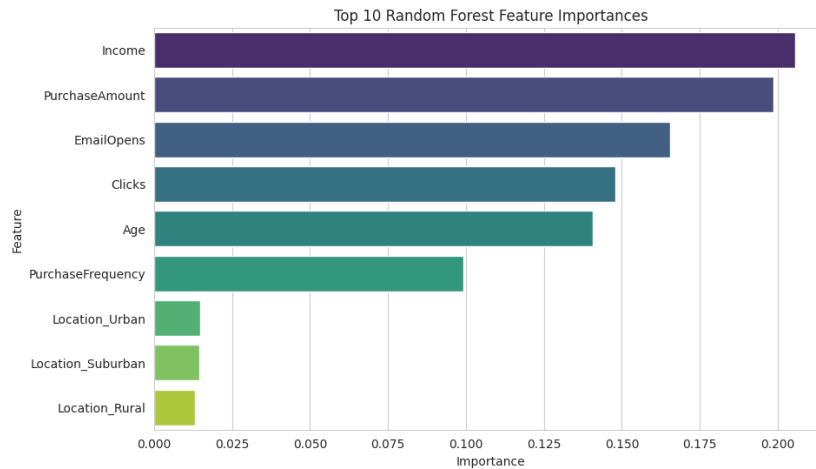


Figure.15. Feature Importance Visualization

The results indicate that income, purchase amount, and engagement metrics are among the most important predictors of churn. However, the confusion matrix reveals that the model still struggles to accurately identify churn cases, highlighting the need for further improvements.

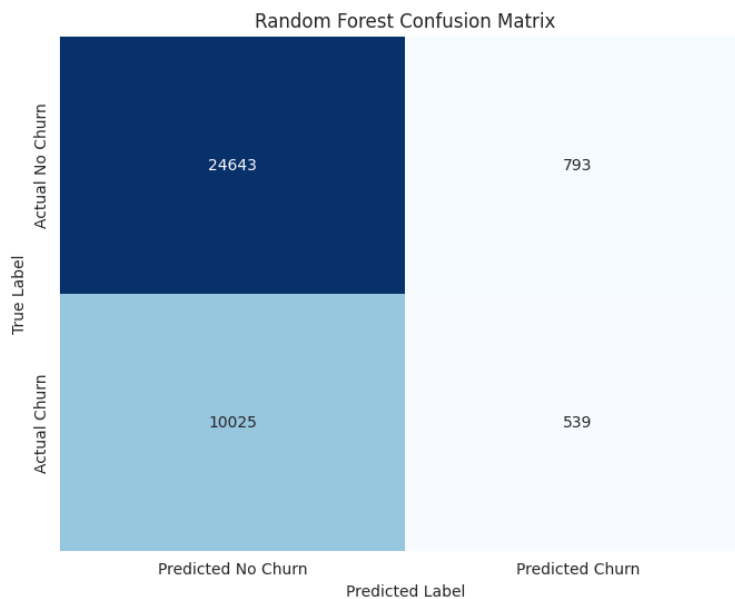


Figure.16. Random Forest Confusion Matrix

The Decision Tree model provides a visual representation of decision rules, showing that purchase frequency is the most critical factor influencing churn.

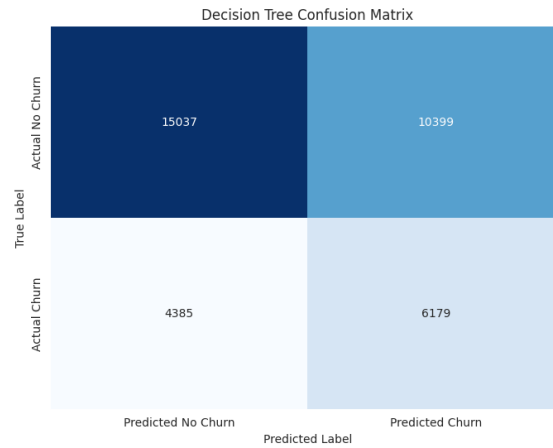


Figure.17. Decision Tree Confusion Matrix

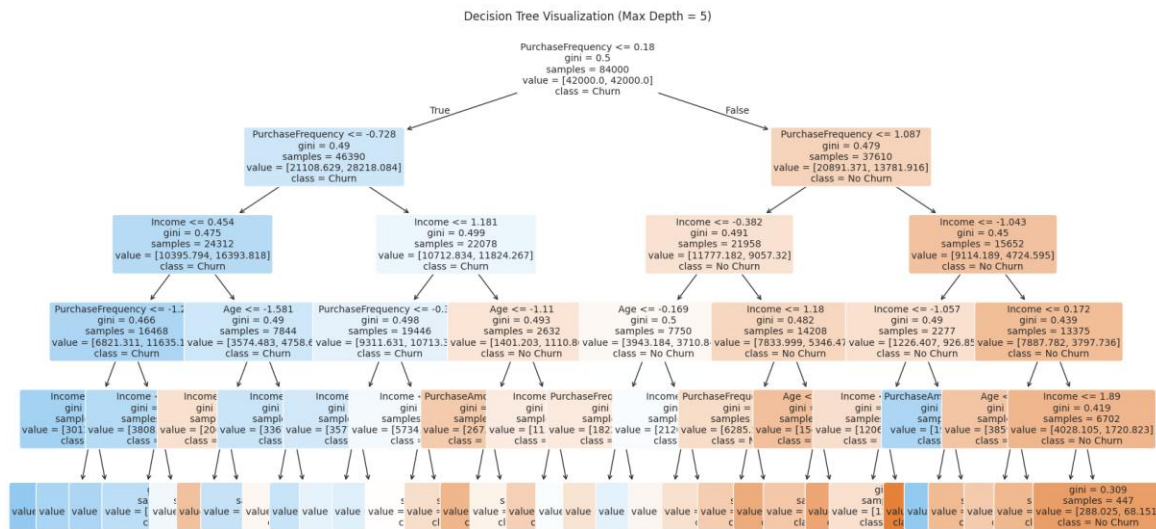


Figure.18. Decision Tree Visualization

3.10 Feature-Level Analysis of Churn Behaviour

A detailed analysis of individual features reveals that purchase frequency is the most significant factor differentiating churn and non-churn customers. Customers with higher purchase frequency exhibit lower churn rates, emphasizing the importance of consistent engagement.

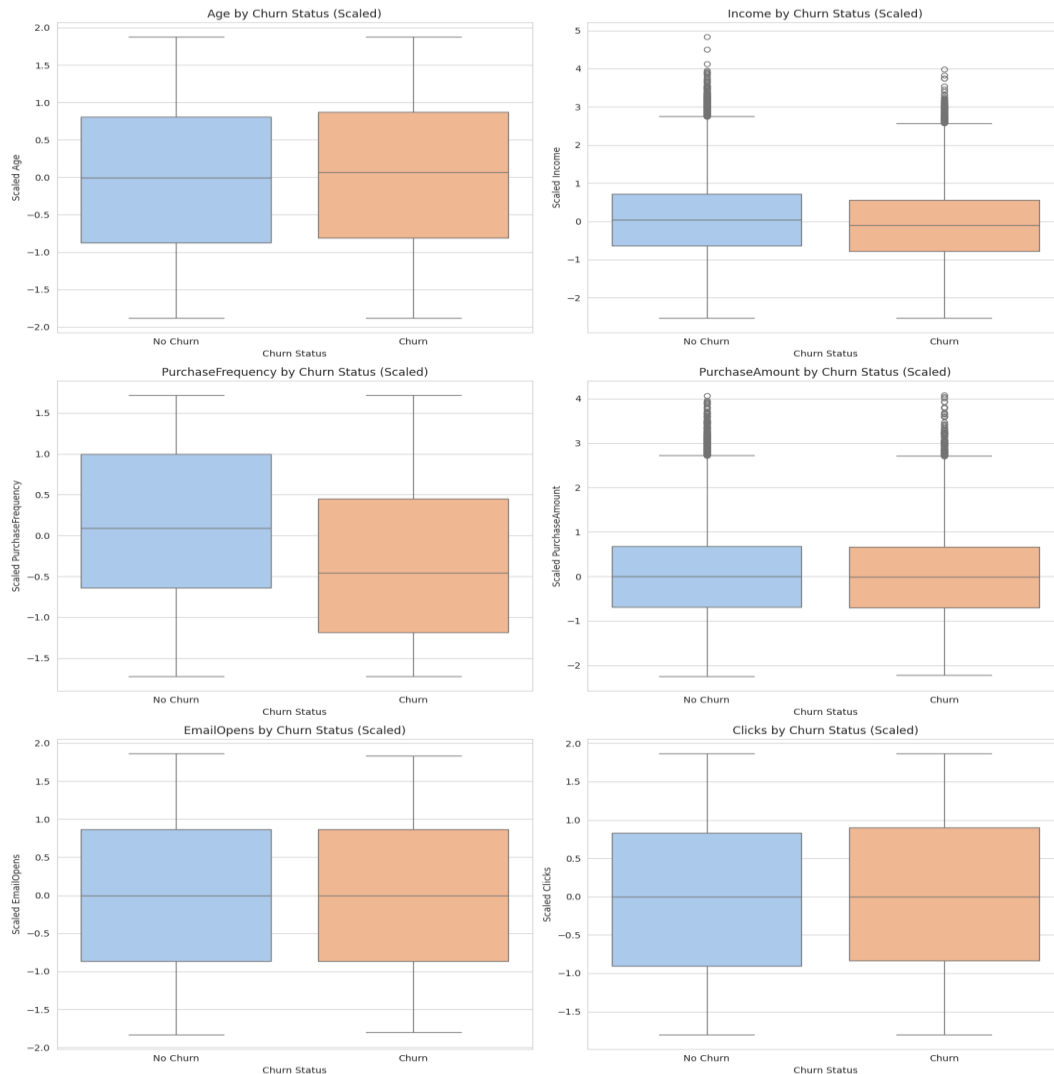


Figure.19. Feature vs Churn Analysis

Other variables, such as income and purchase amount, also show some variation, while engagement metrics display overlapping distributions, indicating limited predictive power.

CHAPTER 4: FINDINGS, CONCLUSION, AND RECOMMENDATIONS

4.1 Key Findings from the Study

The study reveals several important insights into customer behaviour and CRM effectiveness. Purchase frequency emerges as the most critical factor influencing customer retention, with higher interaction levels leading to significantly lower churn rates. Income also plays a notable role, with higher-income customers demonstrating greater loyalty.

The analysis further highlights the importance of early customer engagement, as the initial phase of the customer lifecycle is crucial in determining long-term retention. Additionally, machine learning models provide valuable predictive capabilities, enabling the identification of at-risk customers.

Customer segmentation reveals distinct behavioral patterns, allowing for targeted marketing and personalized engagement strategies.

4.2 Conclusion on CRM Effectiveness

The CRM system implemented by Ford Motor Company is highly effective in managing customer relationships and driving business growth. By integrating advanced technologies such as artificial intelligence, predictive analytics, and connected vehicle data, Ford has successfully transformed its CRM approach into a strategic organizational capability.

The company's focus on personalization, omni-channel engagement, and proactive customer interaction has significantly enhanced customer satisfaction and loyalty. The ability to predict churn and identify high-value customers further strengthens its competitive advantage. Overall, Ford's CRM strategy demonstrates how data-driven decision-making can lead to improved customer retention and increased profitability.

4.3 Recommendations for Improving CRM Practices

Despite its strengths, there are opportunities to further enhance Ford's CRM system. One key recommendation is to improve churn prediction accuracy by incorporating advanced techniques such as deep learning and handling class imbalance more effectively.

Additionally, the company should focus on increasing engagement among low-frequency customers through targeted campaigns and personalized incentives. Enhancing the FordPass application with AI-driven features such as chatbots and predictive recommendations can further improve user experience.

Another important recommendation is to develop flexible pricing and financing options for price-sensitive customers, particularly in lower-income segments. Gamification of loyalty programs can also be introduced to increase customer engagement and interaction.

4.4 Final Conclusion

In conclusion, the integration of CRM practices with data analytics and digital technologies has enabled Ford to build strong and lasting customer relationships. The insights derived from this study highlight the importance of personalization, early engagement, and predictive modelling in enhancing CRM effectiveness.

By continuously refining its strategies and adopting innovative technologies, Ford can further strengthen its position as a customer-centric organization and achieve sustainable long-term growth.

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